Chapter 33

Advanced Clinical Pharmacy Practitioner

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Learning Objectives:

Objective 33.1 Review the impact of pharmacist-provided interventions.
Objective 33.2 Overview innovative pharmacy practice models.
Objective 33.3 Explain the development of competencies in pharmacy workforce leadership.
Objective 33.4 Detail continuing education needs of pharmacists.

OBJECTIVE 33.1. REVIEW THE OUTCOMES OF PHARMACIST-PROVIDED INTERVENTIONS

Pharmacists are versatile members of the healthcare team. Residency training is becoming a common method for pharmacists to advance their skills and knowledge following pharmacy school,[1] allowing pharmacists to be increasingly involved throughout the medication use process. This trend continues from a long history of improving patient care. In one of the most widely cited reviews of pharmacist impact on patient care, the 2011 Report to the US Surgeon General described published studies noting pharmacist services having a positive impact on healthcare access, reduction of healthcare disparities, and improved outcomes with diabetes, hypertension, dyslipidemia, congestive heart failure, and cost-effectiveness of medication use.[2] These benefits and many others have been described in all pharmacy practice areas. An initiative that led to 80% of nursing homes in Singapore choosing to benefit from pharmaceutical care was jointly awarded the 2018 Pharmacy Practice Improvement Award by the International Pharmaceutical Federation (FIP). The award, given to the Pharmaceutical Society of Singapore (PSS), was shared with the American Pharmacists Association (APhA) for a model of care that maximizes the opportunity for pharmacists to contribute to vaccine coverage.[37]

Inpatient Practice

Pharmacists practicing in hospitals are commonly involved in medication dispensing, medication reconciliation, medication order review, monitoring, dosing, and patient counseling (Fig. 33.1).[3] Pharmacists are consistent members of medical rounding teams, where therapeutic decisions are discussed and implemented. Participation on rounding teams has been shown to decrease adverse drug events, with the potential to reduce hospital length of stay.[3–5] Through medication dosing and order verification practices, pharmacists improve both international normalised ratio (INR) control with warfarin therapy and dosing of vancomycin and aminoglycosides.[3] Pharmacist involvement in antimicrobial stewardship programs aims to improve antibiotic selection, dosing, and duration of therapy, reduce the emergence of antibiotic resistance, and decrease costs associated with antimicrobial use.[6] Medication reconciliation and discharge counseling

Clinical Pharmacy Education, Practice and Research. https://doi.org/10.1016/B978-0-12-814276-9.00033-7
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FIGURE 33.1 Patient counseling at bedside.

have been shown to decrease the risk of hospital readmission within 30 days of discharge.[7–9] In some cases, pharmacy services have even been associated with decreased hospital mortality rates.[10]

Ambulatory Practice

Patients spend the majority of their time in ambulatory clinic practice settings. Pharmacists in these settings, like their counterparts in inpatient practices, can work closely with the medical team to devise and adjust short- and long-term therapy plans for a wide range of conditions. By managing medication dosing and monitoring, pharmacists have improved hypertension and diabetes control, improving the quality of life for patients.[11–17] In many cases, pharmacist interventions leading to improved quality also lead to decreased overall costs of care.[11,15,16] Pharmacists are also commonly involved in medication adherence initiatives, utilizing motivational interviewing and other techniques to improve patients' abilities to use prescribed medications effectively.[16] Additionally, pharmacists have successfully decreased the unnecessary use of medications, especially in elderly patients.[17] With demonstrated benefits in a wide variety of clinical conditions, pharmacists have numerous opportunities to improve care in ambulatory settings.

Community Pharmacy Practice

Community pharmacy practice comprises the majority of licensed pharmacists, wherein pharmacists review the appropriateness of prescribed medications as part of the dispensing process for ambulatory patients. Also, various other cognitive clinical services can be performed that can be funded by the patient, health insurance, or government, depending on the country. Given their availability, many surveys of patient and physician perceptions of pharmacist roles and impact are driven by services provided by community pharmacists. Clinical interventions in community practice settings often focus on comprehensive medication use reviews and patient counseling to decrease polypharmacy, improve dosing, and improve patient adherence. These programs are collectively referred to as Medication Management Services, officially recognized in the United States by the Centers for Medicare and Medicaid Services as a reimbursable pharmacist-provided service in community pharmacies. Similar recognition is provided in other countries. Medication therapy management programs, which may include pharmacist prescribing, identify drug therapy problems (such as needing additional medication therapy to treat an established condition) leading to improved disease control, and cost savings for health systems.[18]

OBJECTIVE 33.2. OVERVIEW INNOVATIVE SPECIALTY PHARMACY PRACTICE MODELS

With successful demonstration of improved patient care outcomes, pharmacists are a valuable resource for health systems. In the United States, a pharmacist's scope of practice (what they are legally allowed to do as a licensed pharmacist) is determined by each state. As such, how a pharmacist interacts with patients and medical teams, including the degree of autonomy granted, may differ by practice location. Related to a pharmacist's scope of practice is the ability to receive payment for some types of nondispensing pharmacy services, based in some cases on whether the pharmacist is deemed a healthcare provider and thus has "provider status" and is allowed to submit for reimbursement of services provided.

Prescriptive Authority

As described previously, pharmacists have been shown to improve patient care quality. As their interventions rely on the knowledgeable use of medications, prescriptive authority is a natural progression for pharmacists with advanced training or demonstration of appropriate skills and knowledge. Many types of prescriptive authority exist for pharmacists.[19]

Policy- or protocol-driven prescribing allows a limited amount of action by the pharmacist based on previously approved clinical decisions. Examples include formulary interchanges in a hospital or initial response to elevated INR in patients receiving warfarin. Decisions are mapped out in detail, and pharmacists assess clinical situations to determine which plan to implement.

A broader application of clinical knowledge is used in collaborative practice, wherein a written agreement outlines criteria for patients the pharmacist may address, responsibilities of the pharmacist, training requirements for the pharmacist, and broad guidelines to ensure consistent care.[20] Under collaborative practice agreements, more flexibility is afforded to pharmacists in determining appropriate actions than is typically seen in protocol-driven prescribing. In the United States, nearly all states allow collaborative practice agreements.[21]

In both protocols and collaborative practice examples, pharmacist prescribing is tied to an agreement with a group of physicians (dependent prescribing). Independent prescribing, as is available in some healthcare systems such as the United Kingdom, does not require prior agreements with physicians to permit assessment and treatment.[22] Independent prescribing is also available to some US practices, such as inpatient services achieved through privileging.[23] As the Centers for Medicare and Medicaid Services defined "medical staff" to apply to pharmacists at the discretion of each hospital, it created an opportunity to apply medical staff bylaws in defining pharmacists' scope of practice and thereby "privilege" a pharmacist to prescribe in some cases. Through privileging, the hospital can review the credentials of nonphysician clinicians and permit independent action within the scope of practice outlined by the state.[23] Regardless of the method used, prescriptive authority is critical for pharmacists to exercise their clinical expertise to deliver care efficiently.

Pharmacist Providers

Although prescriptive authority provides the ability for pharmacists to practice at a high level, establishing pharmacy services requires some type of financial benefit to offset the salary of the pharmacist (through reimbursement in fee-for-service models or improved clinical outcomes in outcome-driven payment models). Without recognizing pharmacists as care providers, many healthcare institutions may struggle to recognize and support pharmacist roles, opting instead to utilize physicians, nurse practitioners, or physician assistants to meet patient needs easily addressed by pharmacists. Achievement of provider status in the United States is recognized by many pharmacy leaders as a high priority for the profession given its significant impact in supporting advanced pharmacist roles.[24]

OBJECTIVE 33.3. EXPLAIN THE DEVELOPMENT OF COMPETENCIES IN PHARMACY WORKFORCE LEADERSHIP

Evolution of the role of pharmacists and pharmacy technicians continues in nearly all parts of the world. In most settings, there is a shift away from pharmacist involvement in traditional compounding, preparation, and dispensing roles toward practicing as clinical pharmacists in patient care roles, usually as part of the healthcare team. The primary difference between one country and another is the status of this transition. Some countries are well along the pathway and have a significant portion of pharmacist time spent in a clinical role providing patient care, while others might be far more restrictive, just beginning this transition.

The reason for the shift is similar regardless of location: traditional preparation and dispensing roles are replaced by automation and technology and the availability of commercial products, plus the growing role of support personnel such as pharmacy technicians. There is also significant supporting evidence of the value that pharmacists bring as clinical pharmacists in these new roles, improving patient outcomes, safety, and lowering cost. In most US hospitals, in addition to monitoring drug therapy in patients, clinical pharmacists also have total responsibility for managing certain medications. In these cases, the pharmacist is responsible for and has the authority to select and order the initial dosage of medication, order necessary laboratory tests associated with the medication, and then to adjust the dosage and frequency of the medication based on the laboratory results. Examples of medications where this commonly occurs are vancomycin, aminoglycoside antibiotics, and other medications that require dosage adjustment based on the renal status of the patient. In ambulatory clinics, pharmacists perform similar roles, working more independently and receiving patient referrals from physicians. Examples of where pharmacists manage patients in this setting include anticoagulation clinics (managing warfarin therapy) and medication therapy management clinics.

These changes have a profound impact on undergraduate and postgraduate pharmacist training and how they maintain their competence once in practice. Pharmacy education has shifted in many locations toward a much more clinically focused curriculum, with less emphasis on preparations, compounding, and manufacturing. There has also been more focus on graduate training, such as pharmacy residencies, and demonstrating competence through board certification.

Influential pharmacy leaders spearhead changes in pharmacy practice and training needs. Leadership can be described in many ways. To some, it is focused on providing a compelling vision for pharmacy services that others may aspire to achieve and influencing others to achieve set goals.[25] For others, leadership focuses on demonstrating advanced skills and the impact that inspire others to advance their knowledge and skills. However it is defined, some leadership qualities stand out as critical for all pharmacists, not just those in formal management positions. The American Society of Health-System Pharmacists, for example, believes that leadership in assuring safe and effective medication use is a professional obligation for all pharmacists.[26] Indeed, the changing landscape of healthcare requires individuals capable of identifying new opportunities, ensuring pharmacists are trained and available to meet the opportunities, and that healthcare partners recognize the role pharmacists can play in advancing patient care. There are several leadership qualities, particularly useful for successful pharmacy leaders.

Vision

For many pharmacists, practice is about helping improve the health conditions for patients. For the effective advancement of these goals, a clear and defined purpose that can unify diverse pharmacists' interests in a common direction is essential. To be effective, the vision must be compelling and resonate with the pharmacists' core personal values.[27] It must provide a view of pharmacy practice that encourages continual growth and advancement of pharmacist skills and must be reflected in the daily actions of pharmacists and leaders throughout the organization. Although vision statements are often looked at as management tools, any pharmacist, with or without a formal leadership position, can be the source of a compelling vision if he or she identifies the motivation for pharmacy practice and strives each day to act in a way consistent with that motivation.

Communication Skills and Relationship Building

Advanced communication skills, both written and verbal, are often considered essential skills for pharmacy managers. Having a compelling vision is only effective if that vision can be effectively shared with others to foster alignment of purpose. Furthermore, given the role of pharmacists as part of the healthcare team, effective communication of ideas, concerns, and suggestions to other team members (such as physicians and nurses) is critical for establishing effective clinical practices. In practice, communication skills are needed when making recommendations on rounds, when counseling patients on the use of a new medication, when sharing the results of pivotal research with colleagues, when proposing a new pharmacist-run service to senior leaders, and when teaching students and residents how to improve clinical reasoning. For communication to be effective, pharmacists need to build strong relationships up and down the leadership structure, and across our care teams.[28] The relationships that are built with members of a team will have an impact on credibility, integrity, and overall value of the team. At all levels, formal and informal, pharmacists must develop effective relationships in conjunction with excellent verbal and written communication skills not only to perform core functions but also to provide leadership in medication use within the medical team.

Emotional Intelligence

Emotional intelligence consists of capabilities in self-awareness (how we feel and our feelings' impact on our surroundings), self-management (how we respond to our feelings), social awareness (how we respond to others' emotions), and social skill (how we combine our awareness of others to develop their skills and influence their actions).[29] To lead others, we must first understand ourselves and be conscious of our reactions and the way others perceive us. In healthcare, work is understandably emotional—either in helping others cope with a new diagnosis and identify a care plan or in addressing the stress of complex and challenging decisions made by the healthcare team. As a result, all pharmacists benefit from developing emotional intelligence. In practice, emotional intelligence supports leadership by allowing pharmacists to recognize and empathize with motivators for patients and coworkers. Through that empathy, pharmacists can build on common goals and influence others.

Challenge the Status Quo/Change Management

In pharmacy practice advancement, a team that has become accustomed to the day-to-day work can become blind to new opportunities. For pharmacists, the ability to challenge the status quo by seeking to understand the reasons behind the team's processes is key to effective leadership. As healthcare priorities shift, a need to critically evaluate why services are or are not employed affords the greatest opportunity to adapt to new patient care needs. Revisiting old ideas and questioning current practices, when done to assess applicability to current needs, can be an effective way to identify improvement opportunities. When opportunities are identified, managing the change appropriately for a team is a key component of successful leadership. The ability to convey a change and the reasons for the change to a team is part of this navigation, as well as assisting individuals through letting go of the old and embracing the new. John Kotter outlines the eight steps required for change management in his article "Leading Change: Why Transformational Efforts Fail."[30] In this article, he describes the important steps a leader must take to ensure successful change.

Resilience

Resilience is a necessary leadership skill in multiple ways. For starters, workload in healthcare is often perceived to be increasing—the expectation to do more with less is prevalent as a means to address declining reimbursement relative to costs in the United States. Resilience in some cases is a matter of addressing physical, emotional, and spiritual energy so that pharmacists can address the demands and stresses of the job.[31] Resilience is extremely important in a fast-paced emotional environment to ensure appropriate performance. Stephen Covey addresses this in his classic book "The 7 Habits of Highly Effective People." His seventh habit describes the need to Sharpen the Saw, or take time to focus on oneself to reenergize for the complex work environment.[32] For leaders, resilience also means accepting setbacks, such as denials for new service opportunities, and quickly pivoting to new or remaining opportunities. The ability of pharmacists to maintain focus during busy days and quickly move on from setbacks allows greater flexibility and thus greater success in identifying effective ways to improve patient care.

OBJECTIVE 33.4. DETAIL CONTINUING EDUCATION NEEDS OF PHARMACISTS

Most countries and jurisdictions have a regulatory framework for licensing pharmacists, usually after completing the required training and passing a licensure examination. Many also recognize the importance of the pharmacist maintaining their skills and keeping current with new information over their 40 or more year career. Usually, this comes in the form of a minimum requirement for continuing education to keep and maintain a pharmacist license. It is common that the minimum requirement will be stated as the number of education hours one must receive each year, usually leaving the selection of specific topics, format, and other details to the pharmacist. Although this method may ensure that the pharmacist receives additional education on a regular basis, it has many shortcomings because usually a passive process (listening to a lecture) does not require that the pharmacist demonstrate competence or proficiency after attending the program. Also, there is nothing to prevent the pharmacist from choosing to attend educational programs that may not be related to their professional practice, which defeats the intended purpose and does not help in maintaining competence.

Some locations rely on a different model for continuing education known as Continuing Professional Development or CPD. CPD is a model where the pharmacist performs an ongoing self-assessment and identifies their areas of proficiency and where they have knowledge gaps based on what they do in day-to-day practice. The pharmacist then chooses

educational programs that are targeted at closing educational gaps, and then the CPD process starts all over again. If performed correctly, this more individualized process should result in the desired outcome of improving and maintaining the pharmacist's ongoing competence to practice as a pharmacist.

Regardless of the process and what is legally required, pharmacists should strive to take advantage of the many educational opportunities now available in various formats. It is a fundamental responsibility of being a professional.

Certificate Programs

Certificate programs are another form of continuing education or skill development that usually involves completion of a course or series of courses with a specific focus. The length of certificate programs varies, depending on the area of focus, but is typically at least 15 h of instruction and can be 80 h or more. They are notably different from professional, undergraduate, or postgraduate degree programs because of their limited focus and shorter duration. Certificate programs can be conducted live or online and may or may not involve completion of an examination or project to demonstrate a mastery of skills through the program. Examples of certificate programs in pharmacy include immunization, sterile compounding, antimicrobial stewardship, medication therapy management, pharmacy informatics, medication safety, and others.[33] It is important to distinguish a certificate program from certification.

Certification

Certification demonstrates expertise in pharmacy practice specialties and requires successful completion of an extensive examination validated as a measure of knowledge in an area of practice. Certification also requires having a pharmacy degree and meeting the minimum number of years of practice. Most practice areas available for board certification are clinical specialties, but other nonclinical areas are also developing. There is usually specific continuing education required to maintain certification once it is achieved. In the United States, the Board of Pharmacy Specialties (BPS) is the primary organization providing certification for pharmacists, and more than 20,000 individuals are certified worldwide by BPS in eight specialties. The specialty areas kept increasing. Examples of specialties available in 2018 include ambulatory care pharmacy, cardiology pharmacy, critical care pharmacy, geriatric pharmacy, infectious diseases pharmacy, nuclear pharmacy, nutrition support pharmacy, oncology pharmacy, pediatric pharmacy, pharmacotherapy and psychiatric pharmacy.[34] Many organizations provide review courses to help prepare pharmacists for the certification exam and may also provide approved continuing education to maintain certification.

Residency Training

Pharmacy residency training builds on PharmD education and outcomes to contribute to the development of clinical pharmacists responsible for the medication-related care of patients with a wide range of conditions. The program is available for graduate pharmacists and begins with a general pharmacy practice residency that lasts at least 12 months and is referred to as a Post Graduate Year 1 (PGY1) residency. After the PGY1 residency, the individual may choose to pursue specialty training with an additional year or more in a specialty area, and this is known as a Post Graduate Year 2 residency.[35]

It should be noted that residency-training programs are postgraduate programs and are different from experiential training associated with most pharmacy curricula, such as Advanced Pharmacy Practice Experiences or APPEs. The APPEs usually involve rotations, but as a student who is not yet graduated and is not licensed to practice, their role in providing patient care will be more limited, and they will not be able to practice independently.

Residency programs are designed to meet standards, demonstrating best practices for resident training and provision of pharmacy services. The American Society of Health-System Pharmacists (ASHP) provides accreditation of programs in the United States and around the world that evaluates the residency program and pharmacy services against an established residency standard. In 2018, there are more than 2000 programs that are accredited by ASHP, mostly in the United States, but there is a growing number of other countries. Many consider residency training to be a critical component in advancing the practice, demonstrating a high level of pharmacy services, and in training clinical pharmacists for future practice.[36]

CONCLUSION

The pharmacy profession and pharmacists have advanced to deliver specialized patient care services. These successful healthcare service models advance the profession. The pharmacy workforce needs to demonstrate leadership along with other professional competencies to positively influence patient care. Continuing education, certification and residency, and other higher education are methods to advance professional practice.